Symmetry operation:

Symmetry operation is a movement of a body (molecule) such that after movement it has been carried out every point of the body is coincident with an equivalent point in its original orientation.

center of Symmetry:

If the ~~rotation~~ reflection of all atom in a molecule through a center gives an indistinguishable configuration then the operation is called inversion and the point is called center of symmetry.

plane of Symmetry:

If a reflection of all atom in a molecule through a plane gives an indistinguishable configuration then plane is called plane of symmetry.

Plane of symmetry is classified as

- Horizontal plane of symmetry, σ_h
- Vertical plane of symmetry, σ_v
- Diagonal plane of symmetry, σ_d.

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Rotational axis :

If a rotation of a~~th~~ ~~atom~~ molecule by an angle $\frac{2\pi}{n}$ about an axis gives an indistinguishable configuration, then the axis is called Rotational axis. It is denoted by C_n.

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2014

Answer to the question no - 1. (a).

n-fold axis of Symmetry:

If rotation of a molecule by an angle of $\frac{2\pi}{n}$ about axis gives an indistinguishable configuration then the axis is called n-fold axis of symmetry.

plane of Symmetry:

If a reflection of all atom in a molecule through a plane gives an indistinguishable configuration then the plane is called plane of symmetry.

Plane of symmetry is classified as

(I) horizontal plane of symmetry, σ_h → right angle to the principle axis

(II) vertical plane of symmetry, σ_v → containing principle axis

(III) Diagonal plane of symmetry, σ_d.

* Diagonal plane are vertical planes that bisect the angle between two successive C_2 axis.

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2013

Ans. to the question no. 1.(a)

Symmetry operation:

Symmetry operation is the movement of a body (molecule) so that after movement has been the carried out every point of the body is coincident with an equivalent point in its original orientation.

Center of Symmetry:

If the reflection of all atom in a molecule through ax point or a center gives an indistinguishable configuration, then the operation is called center of symm inversion and the centre is called centre of symmetry.

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2011

Ans. no. 1.(a)

Symmetry element:

Symmetry element is a geometrical entities such as an axis, a point or a plane with respect to which one or more symmetrical operations carried out.

There are five elements of symmetry.

Axis of symmetry, C_n
Center of symmetry, i
Plane of symmetry, σ
Improper axis, S_n
Identity.

Symmetry operation:

Symmetry operation is the movement of a body (molecule) such that after the movement has been carried out every point of the body is coincident with an equivalent point in its original orientation.

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(iii) plane of Symmetry:

If the reflection of all atom in a molecule through a plane given an indistinguishable configuration, then the plane is called plane of symmetry. It is denoted by σ.

Horizontal plane of symmetry
Vertical plane of symmetry
Diagonal plane of symmetry.

(iv) Centre of Symmetry:

If the reflection of all atom in a molecule through a center given an indistinguishable configuration, then the operation is called inversion and the center is called center of symmetry. The center of symmetry is denoted by i.

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Answer no. 1.(a)

Symmetry element:

Symmetry element is a geometrical entities such as an axis, a plane or a point with respect to which one or more symmetrical operations may be carried out.

plane of Symmetry:

If reflection of all atom of a molecule through a plane gives an indistinguishable configuration, then the plane is called plane of symmetry.

Centre of Symmetry:

If a reflection of all atom in a molecule through a centre gives an indistinguishable configuration, then the operation is called inversion and the centre is called centre of symmetry.

n-fold axis of Symmetry: (C_n) Proper rotational axis

If rotation of a molecule by an angle $\frac{2\pi}{n}$ about an axis gives an indistinguishable configuration, then the axis is called n-fold axis of symmetry.

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Improper rotation axis: (S_n):

If a rotation of a molecule by an angle of $\frac{2\pi}{n}$ about an axis followed by a reflection through a plane perpendicular to that axis gives indistinguishable configuration, then the ....

Point group:

Point group or symmetry group is a complete set of symmetry operations of a molecule that discribes the molecule overall symmetry. Point group is represented by C_1, C_n, C_{nv}.

Symmetry operation:-

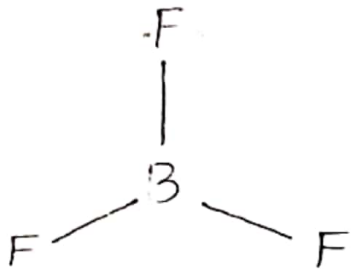
Symmetry operation is the movement of a body such that after the movement has been carried out every point of the body is coincident with an equivalent point in its original orientation.

Symmetry element:-

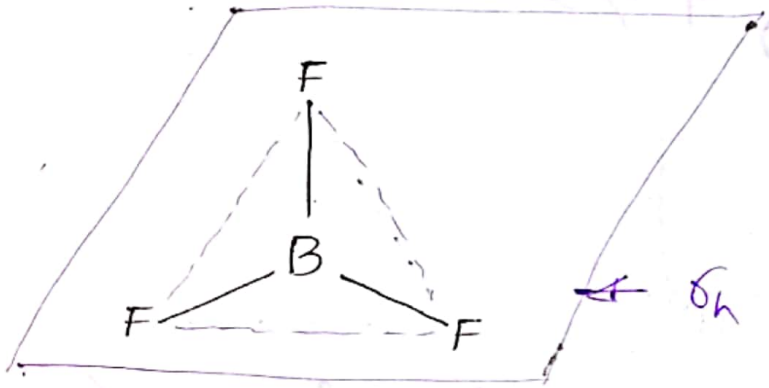
Symmetry element is a geometrical entities such as an axis, a plane or a point with respect to which one or more symmetrical operations carried out.

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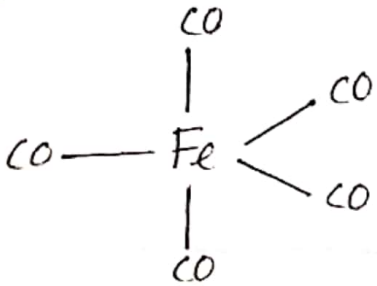
BF_3 → a σ_h in BF_3



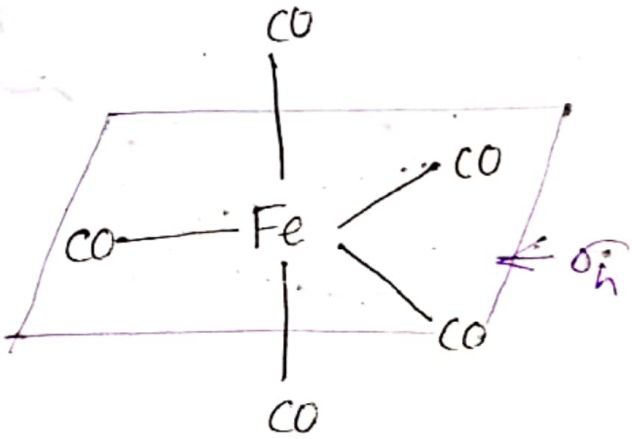
Trigonal planar

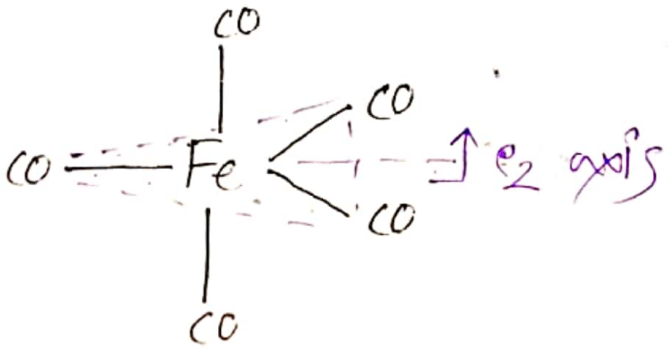


$Fe(CO)_5$ → a C_2 axis and σ_h



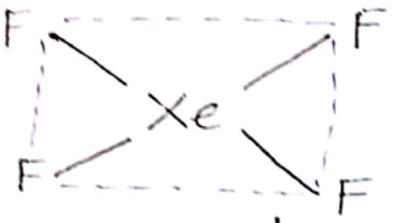
Trigonal bipyramidal



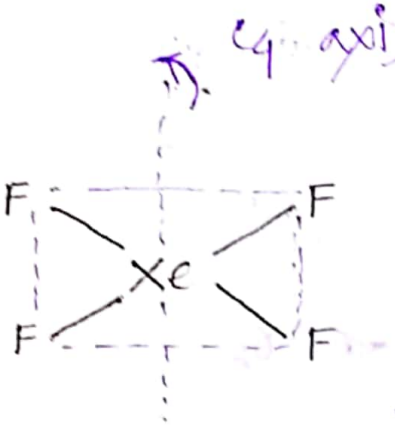


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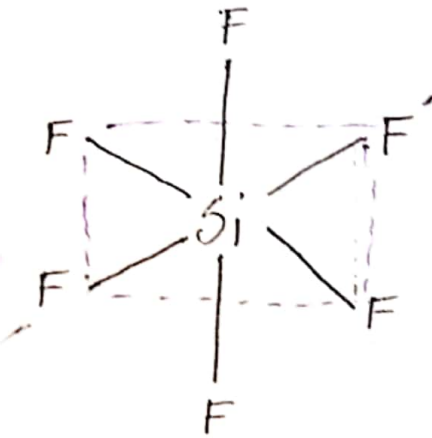
XeF_4 → A C_4 axis and a σ_h

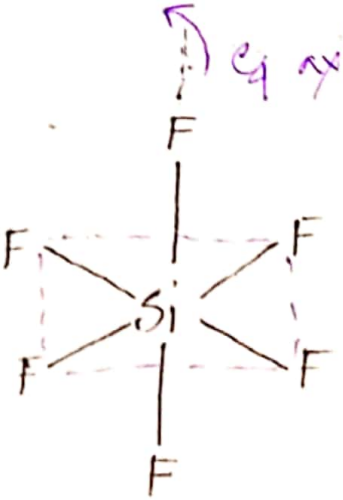


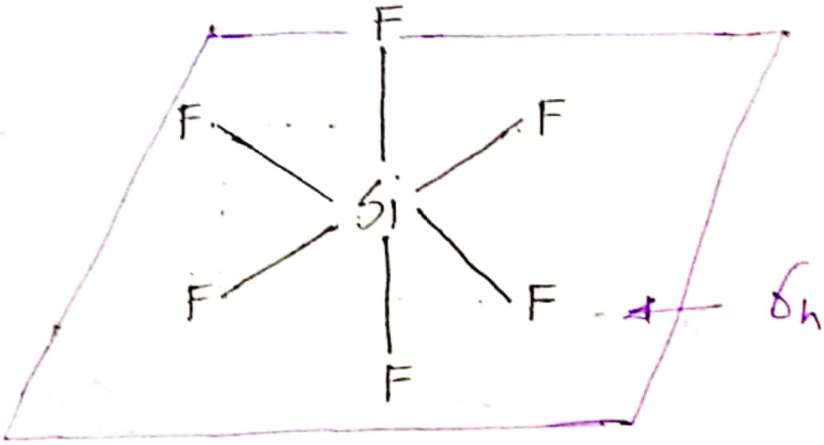
Square planar



SiF_6 → A C_2 axis and a C_4 axis, a σ_h



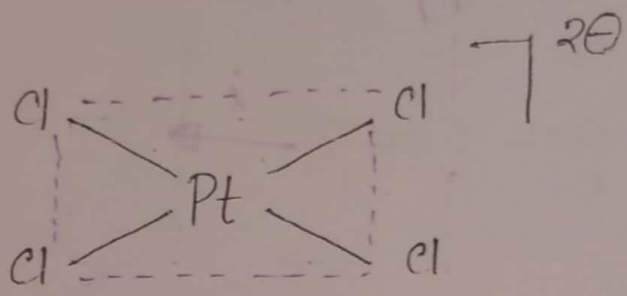




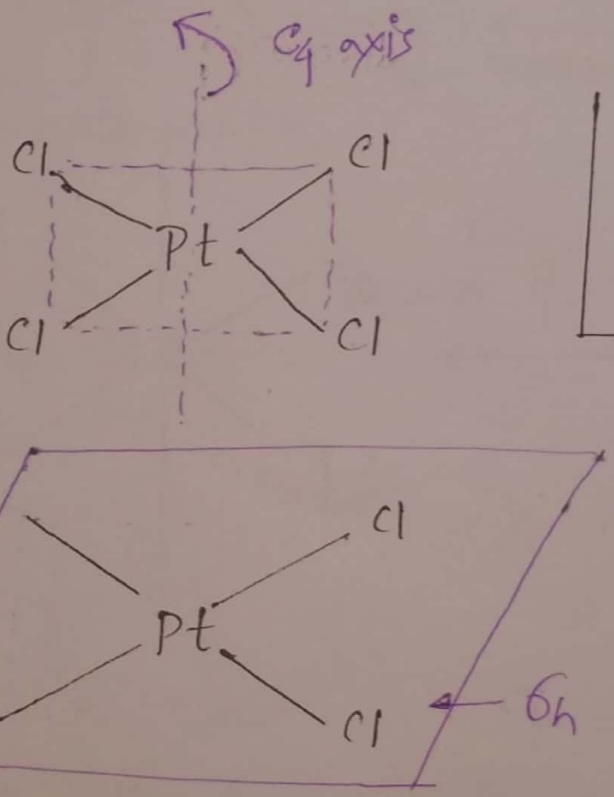
Octahedral

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$[PtCl_4]^{2-}$ → C_4 and σ_h

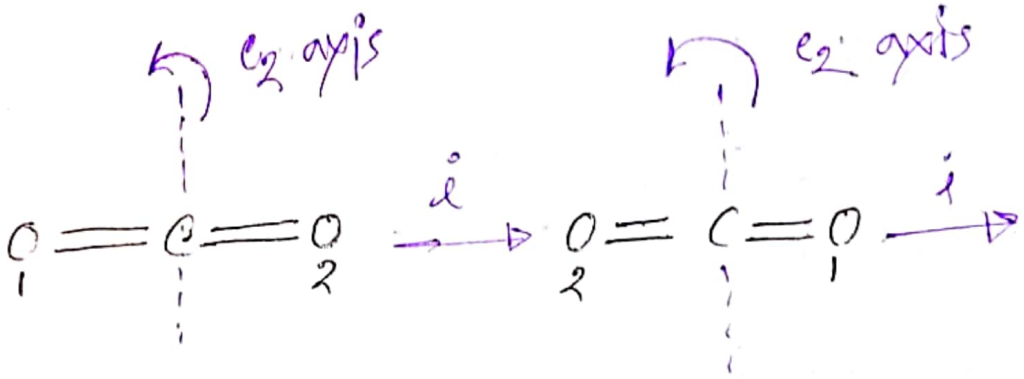


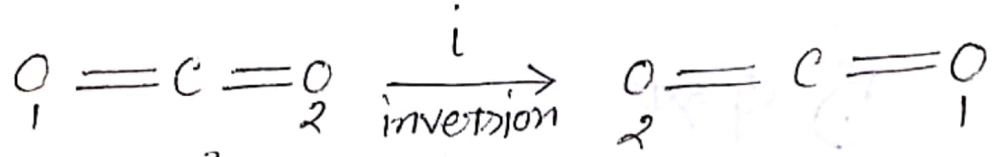
Square planar

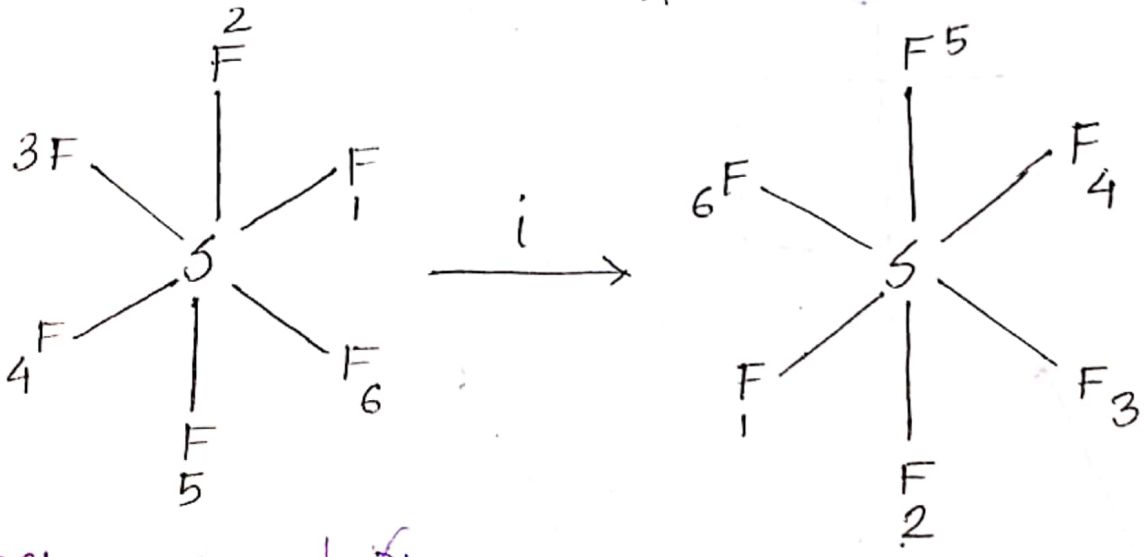


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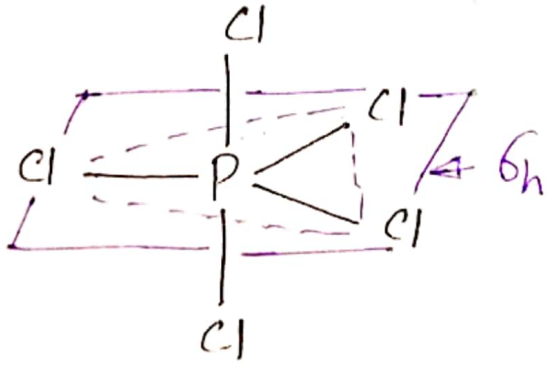
CO_2 → centre of inversion

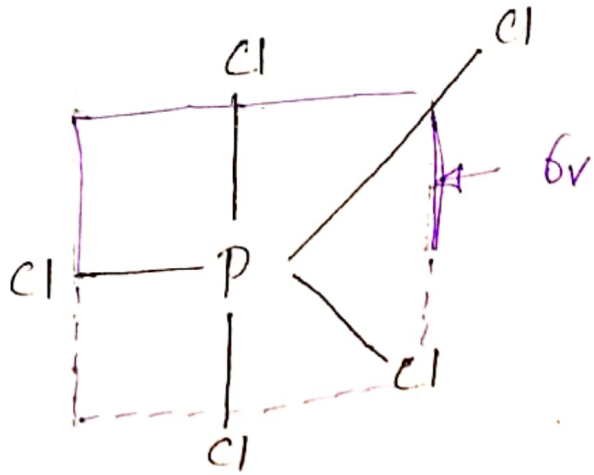






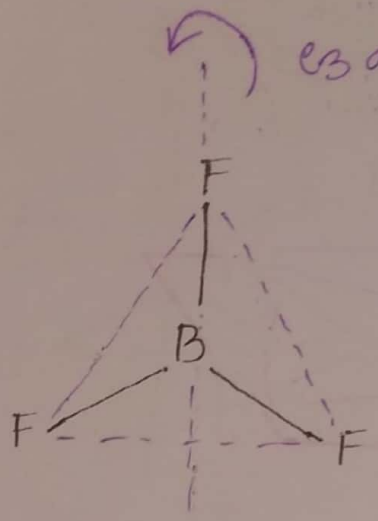
PCl_5 → σ_v and σ_h



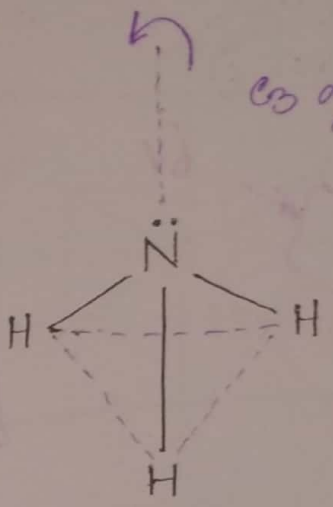


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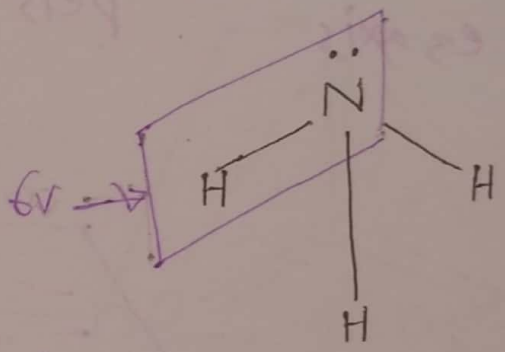
BF_3 → a C_3 axis

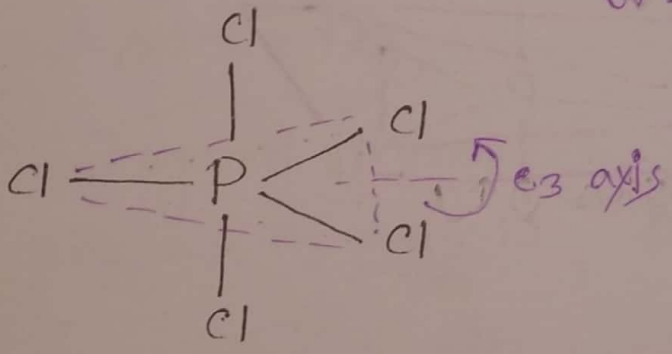


NH_3 → a C_3 axis and σ_v

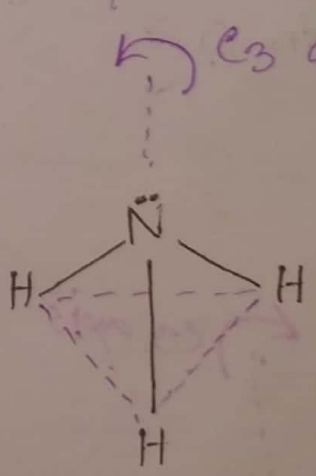


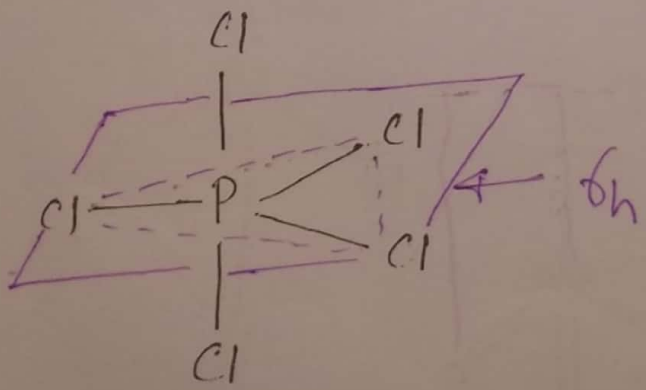
PCl_5 → C_2 axis and σ_h





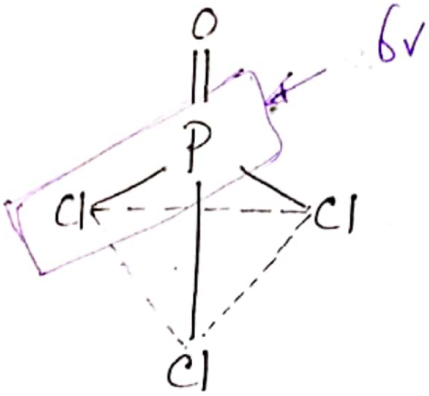
A C_3 axis in NH_3



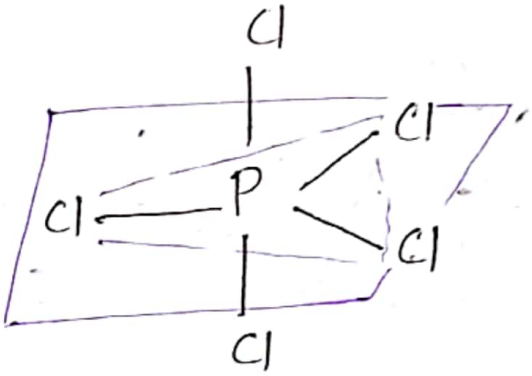


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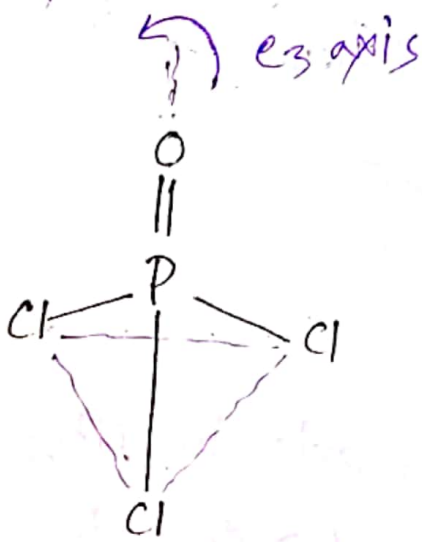
$POCl_3 \rightarrow \sigma_v$



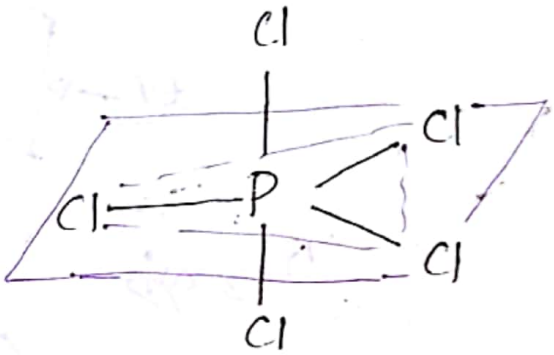
PCl_5

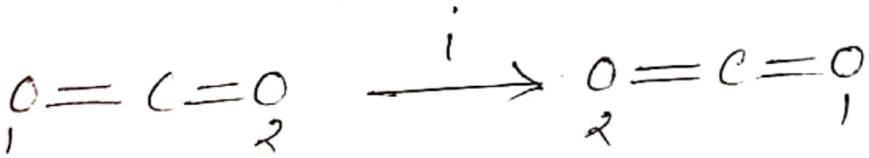


$POCl_3 \rightarrow C_3$

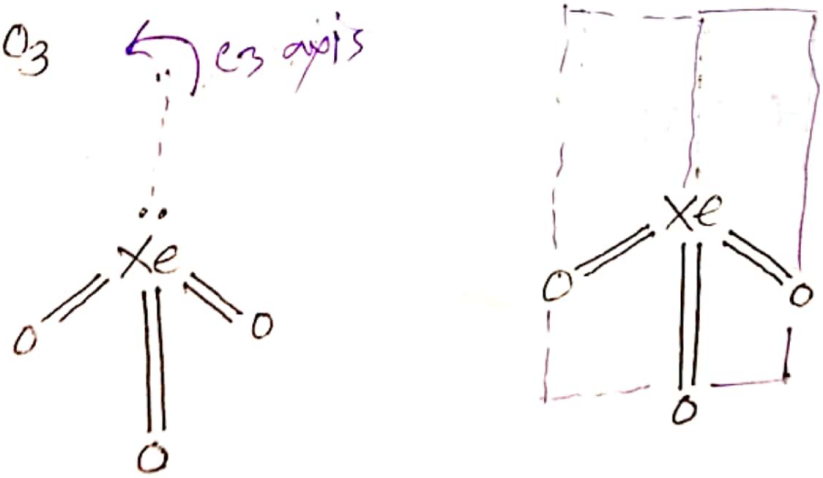


$PCl_5 - 6h$



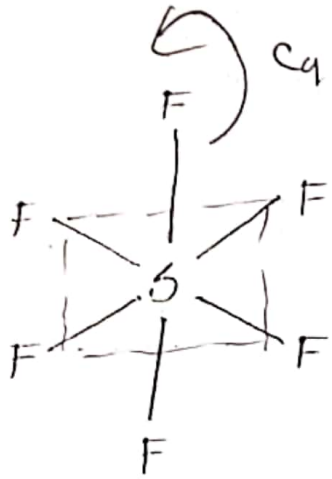


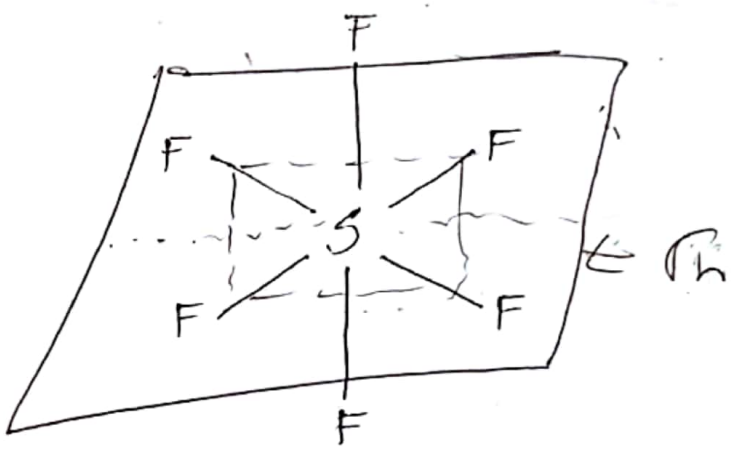
XeO_3



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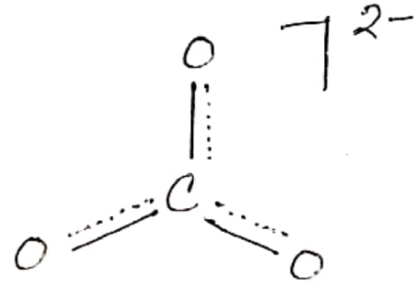
SF_6 → C_4, σ_h



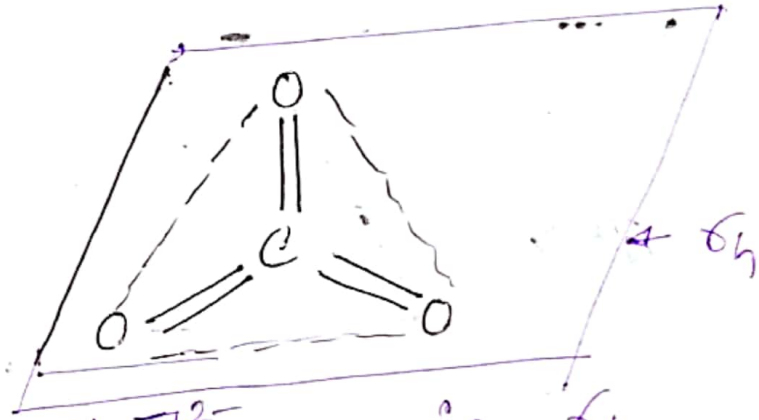


Tear form

CO_3^{2-} → σ_h, $6\sigma_v$

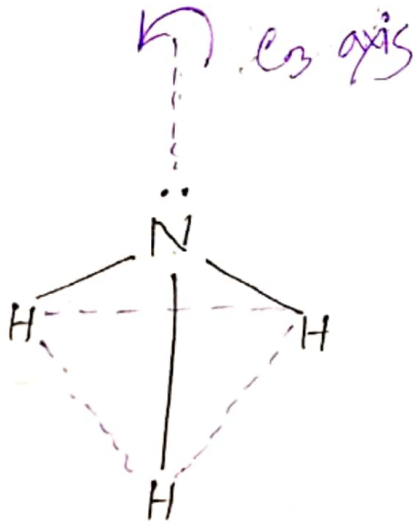


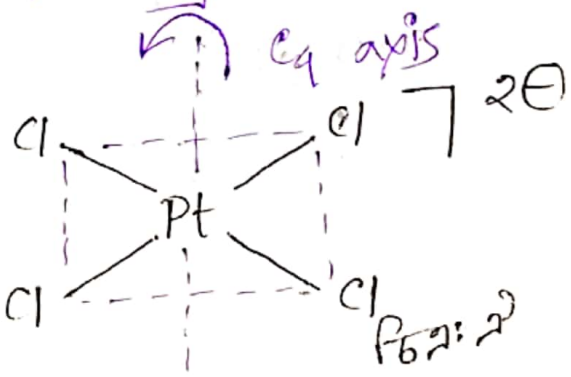
Trigonal planar



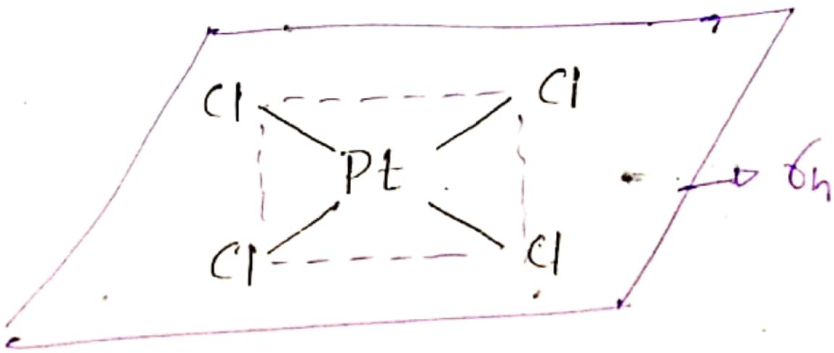
$PtCl_4]^{2-}$ → C_4 axis, σ_h

NH_3 → C_3 axis





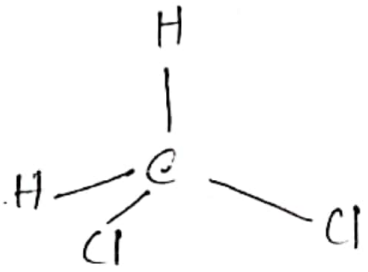
Tear: 2

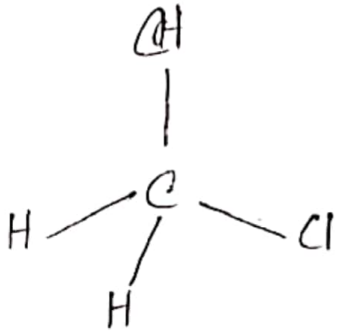


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Point group :

CH_2Cl_2





$C_n = ?$
yes, $n=2$

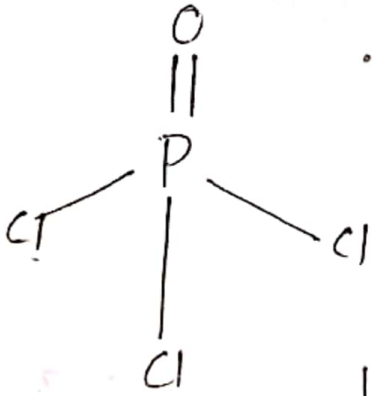
$2C_2 \perp C_2$

σ_v
yes

No

C_{2v}

$POCl_3$



$C_n = ?$
yes,
$n=3$

$3\sigma_v$
yes

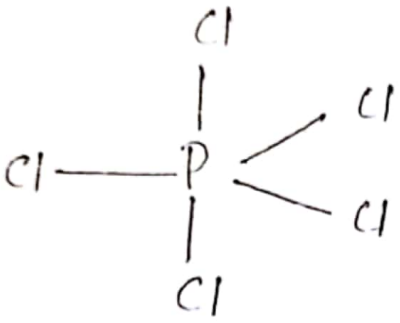
$3C_2 \perp C_3$

No

C_{3v}

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PCl_5



$C_n = ?$ yes $n = 3$

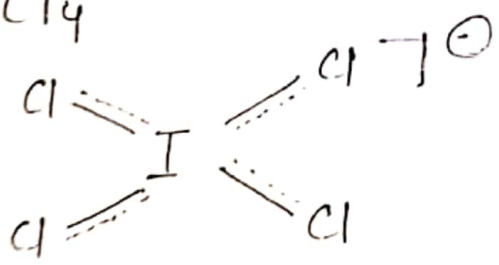
$3\sigma_v$ yes

$3C_2 \perp C_3$ yes

σ_h yes

D_{3h}

ICl_4^-



$C_n = ?$ yes $n = 4$

$4\sigma_v$ yes

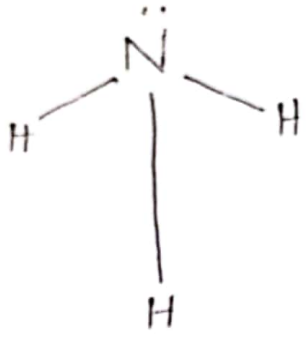
$4C_2 \perp C_4$ yes

σ_h yes

D_{4h}

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NH_3



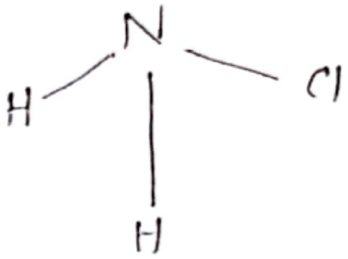
$C_n = ?$ yes $n = 3$

$3\ C_2 \perp \sigma_v$ yes

$3C_2 \perp C_3$ No

C_{3v}

NH_2Cl



$C_n = ?$ No

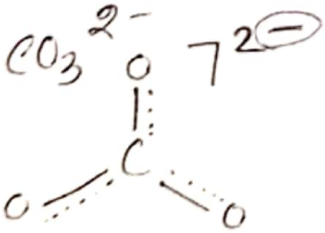
σ

$C_n = ?$ No

σ yes

C_s

CO_3^{2-}



$C_n = ?$ yes $n = 3$

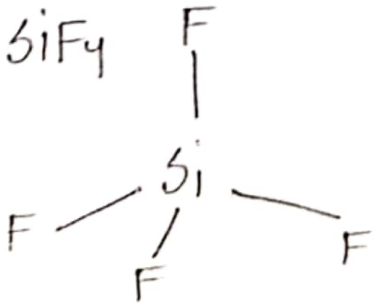
$3\sigma_v$ yes

$3C_2 \perp C_3$ yes

σ_h yes

D_{3h}

SiF_4

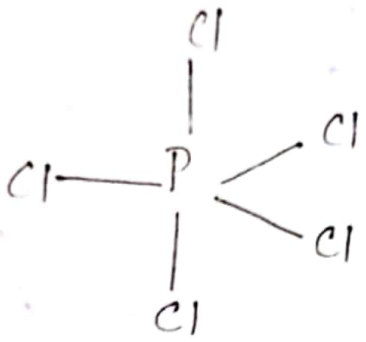


Tetrahedral

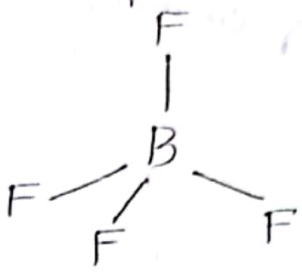
T_d

Scanned with CamScanner

PCl_5



BF_4^-



Tetrahedral

T_d

$C_n = ?$ yes $n=3$

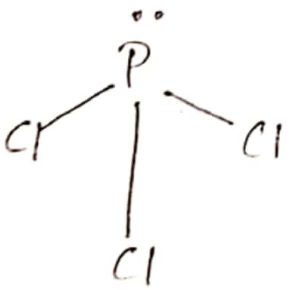
$3\sigma_v$ yes

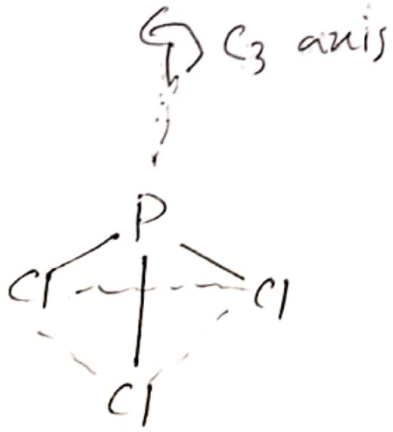
$3C_2 \perp C_3$ yes

σ_h yes

D_{3h}

PCl_3





$C_n = ?$ yes $n=3$

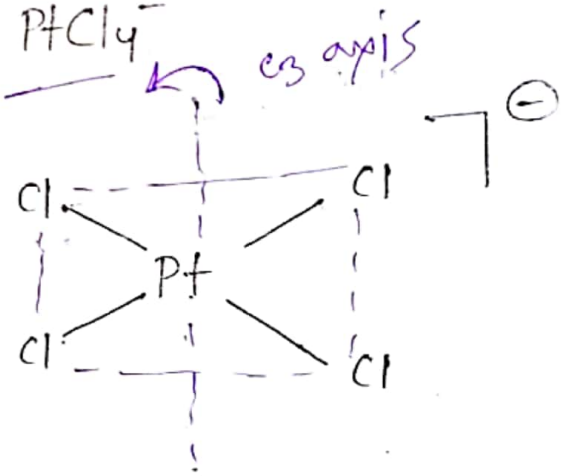
$3\sigma_v$ yes

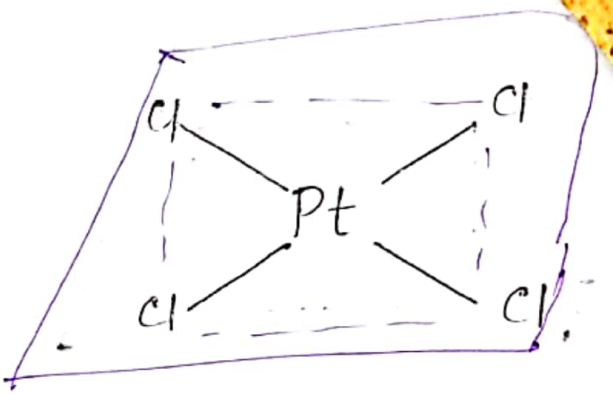
$3C_2 \perp C_3$ No

C_{3v}

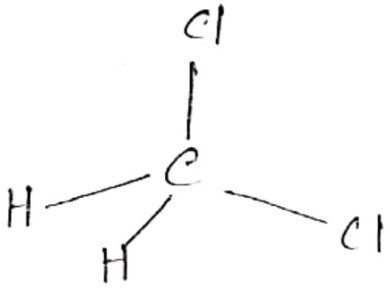
Scanned with CamScanner

$PtCl_4^-$

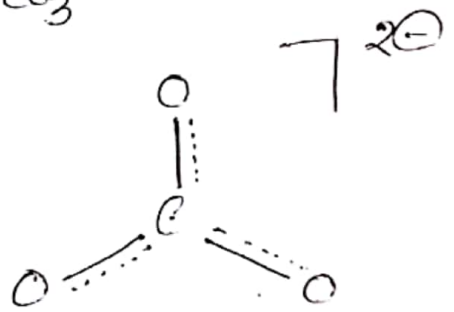


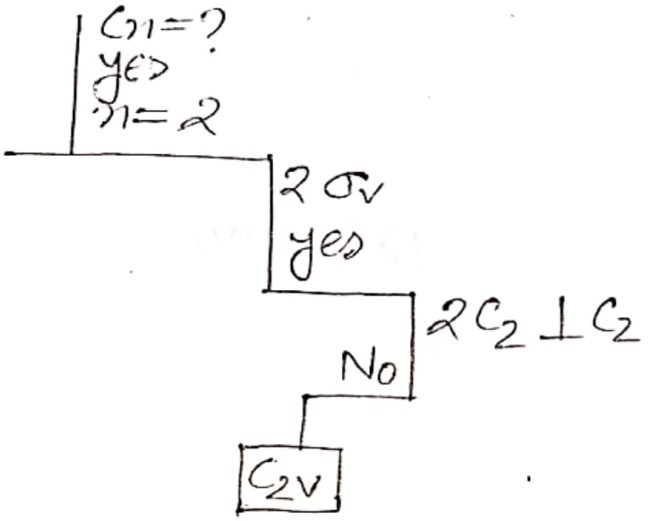


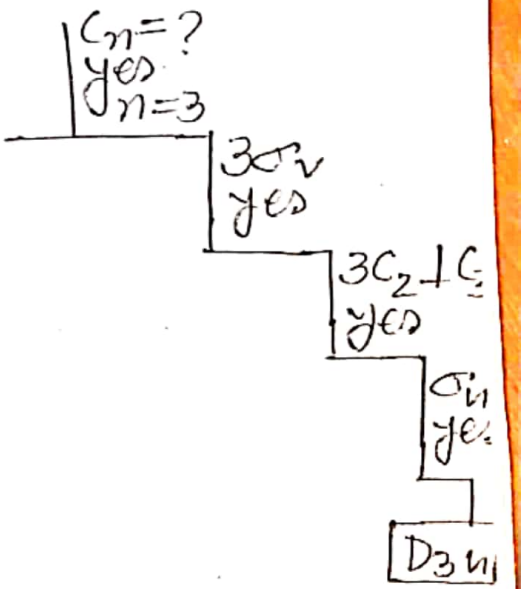
CH_2Cl_2



CO_3^{2-}

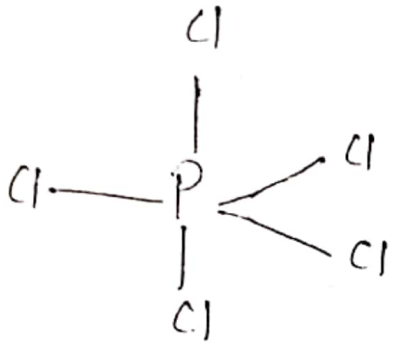






Scanned with CamScanner

PCl_5



$C_n = ?$ yes $n = 3$

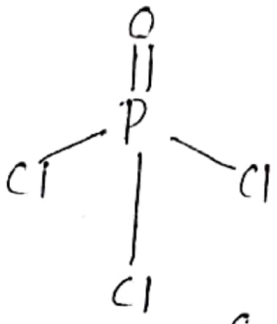
$3\sigma_v$ yes

$3C_2 \perp C_3$ yes

σ_h yes

D_{3h}

$POCl_3$



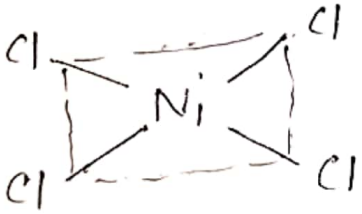
$C_n = ?$ yes $n = 3$

$3\sigma_v$ yes

$3C_2 \perp C_3$ No

C_{3v}

$NiCl_4^{2-}$



$C_n = ?$ yes $n = 4$

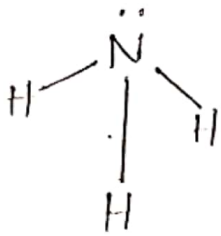
$4\sigma_v$ yes

$4C_2 \perp C_4$ yes

σ_h yes

D_{4h}

NH_3



$C_n = ?$ yes $n = 3$

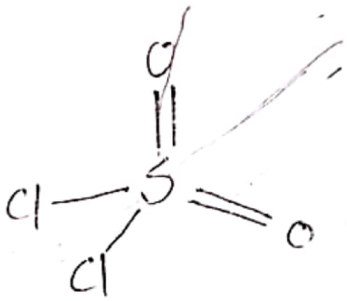
$3\sigma_v$ yes

$3C_2 \perp C_3$ No

C_{3v}

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SO_2Cl_2



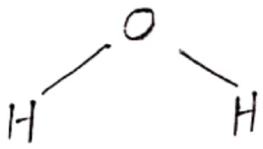
$C_n = ?$ yes $n = 2$

$2\sigma_v$ yes

$2C_2 \perp C_2$ No

C_{2v}

H_2O



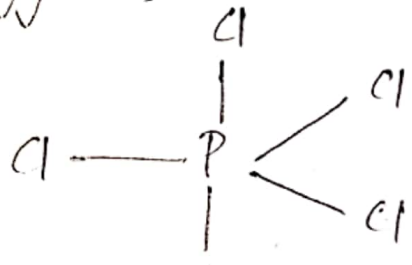
$C_n = ?$ yes $n = 2$

$2\sigma_v$ yes

$2C_2 \perp C_2$ No

C_{2v}

PCl_5



$C_n = ?$ yes $n = 3$

$3\sigma_v$ yes

$3C_2 \perp C_3$ yes

σ_h yes

D_{3h}

C_6H_6



$C_n = ?$ yes $n = 6$

$6\sigma_v$ yes

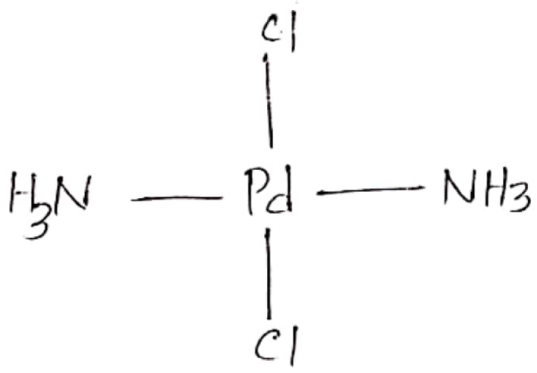
$6C_2 \perp C_6$ yes

σ_h yes

D_{6h}

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trans - $[Pd(NH_3)_2Cl_2]$



$C_n = ?$ yes $n = 2$

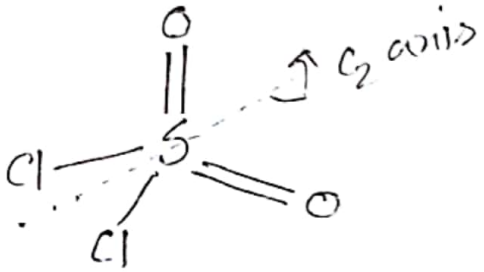
$2\sigma_v$ yes

$2C_2 \perp C_2$

No

C_{2v}

SO_2Cl_2



$C_n = ?$ yes $n = 2$

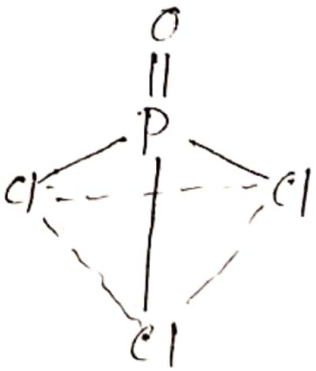
$2\sigma_v$ yes

$2C_2 \perp C_2$

No

C_{2v}

$POCl_3$



$C_n = ?$ yes $n = 3$

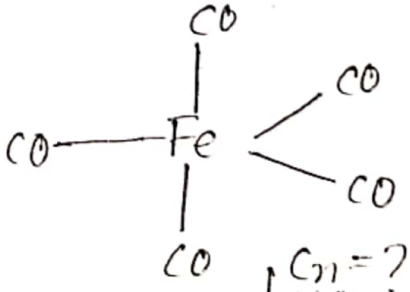
$3\sigma_v$ yes

$3C_2 \perp C_3$

No

C_{3v}

$Fe(CO)_5$



$C_n = ?$ yes $n = 3$

$3\sigma_v$ yes

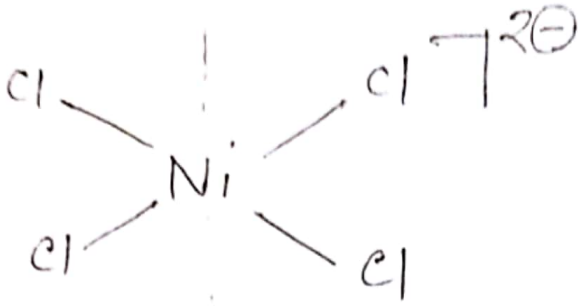
$3C_2 \perp C_3$ yes

σ_h yes

D_{3h}

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$NiCl_4^{2-}$



$C_n = ?$ yes $n = 4$

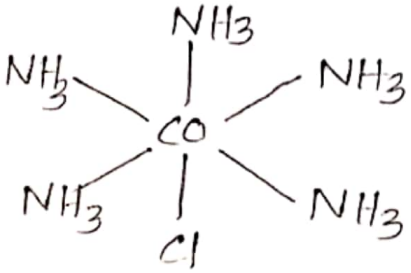
$4\sigma_v$ yes

$4C_2 \perp C_4$ yes

σ_h yes

D_{4h}

$COCl(NH_3)_5$



$C_n = ?$ yes $n = 4$

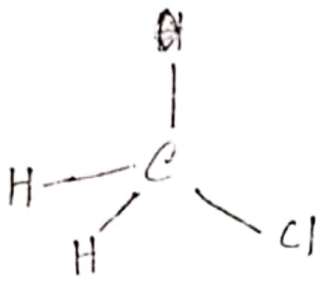
$4\sigma_v$ yes

$4C_2 \perp C_4$ No

C_{4v}

Scanned with CamScanner

CH_2Cl_2



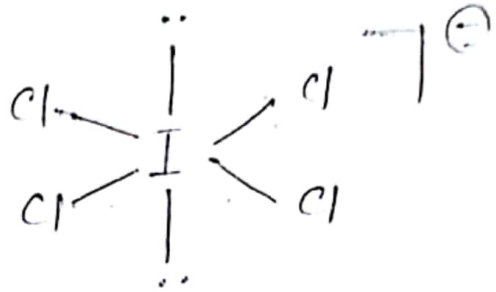
$C_n = ?$ yes $n = 2$

$2C_2$ yes

$2C_2 \perp C_2$ No

C_{2v}

ICl_4^-



$C_n = ?$ yes $n = 4$

$4\sigma_v$ yes

$4C_2 \perp C_4$ yes

σ_h yes

D_{4h}

CH_4 is a regular tetrahedral molecule, so the point group of CH_4 is T_d

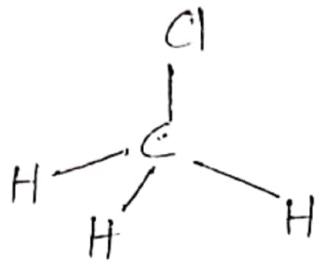
SF_6 is a regular octahedral molecule, so the point group is O_h.

HCl is a assymmetrical linear molecule so the point group is $C_{\infty v}$

H_2 is a symmetrical linear molecule, so the point group is $D_{\infty h}$.

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CH_3Cl



$C_n = ?$ yes $n = 3$

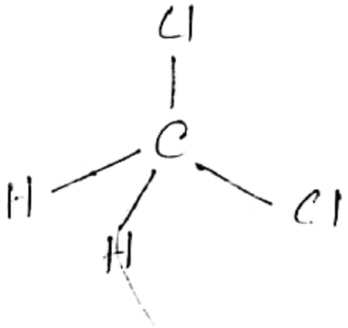
$3\sigma_v$ yes

$3C_2 \perp C_3$

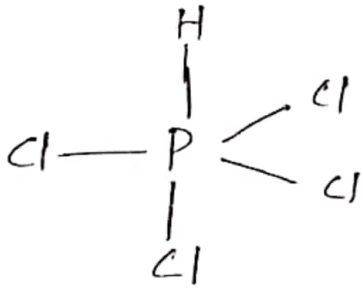
No

C_{3v}

CH_2Cl_2



PCl_4H



$C_n = ?$ yes $n = 3$

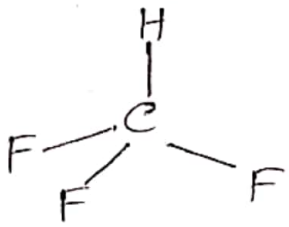
$3\sigma_v$ yes

$3C_2 \perp C_3$

No

C_{3v}

CHF_3



$C_n = ?$ yes $n = 3$

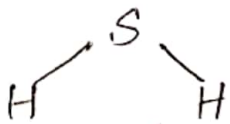
$3\sigma_v$ yes

$3C_2 \perp C_3$

No

C_{3v}

H_2S



$C_n = ?$ yes $n = 2$

$2\sigma_v$ yes

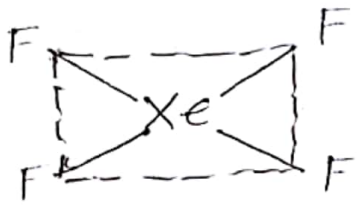
$2C_2 \perp C_2$

No

C_{2v}

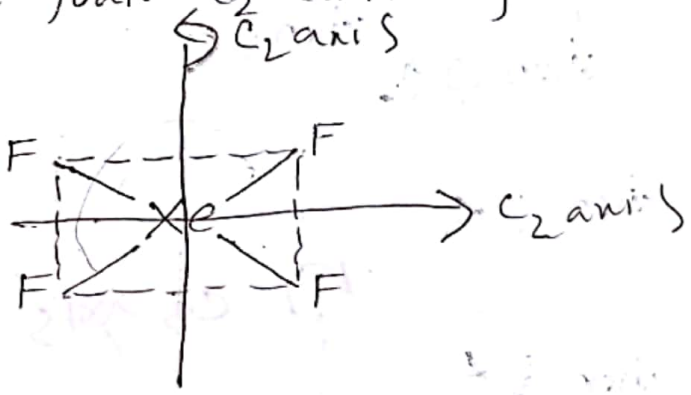
Scanned with CamScanner

XeF_4



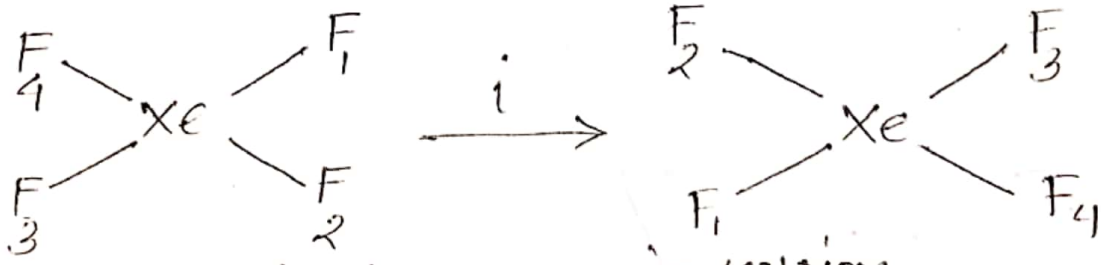
Square planar

(I) There are four C_2 axis of rotation.



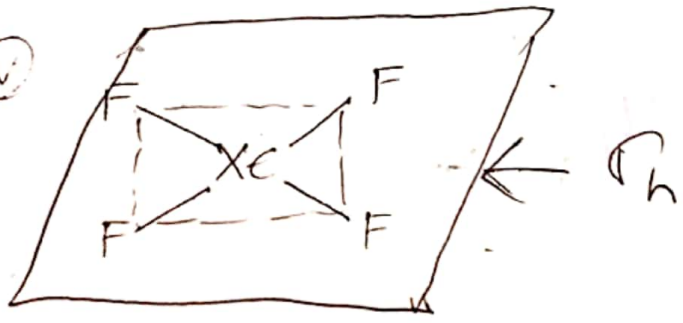
(II) There is no C_3 axis of rotation.

(III)



The molecule has an inversion.

(IV)

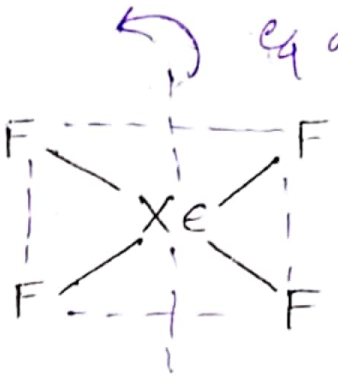


Scanned with CamScanner

(v) If a molecule has a center of inv
then the molecule can not be polar.

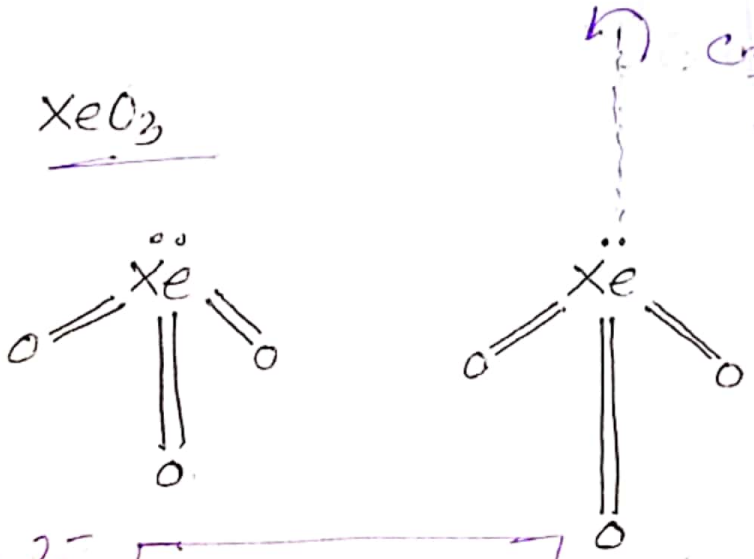
Since the XeF_4 molecule has center of inversion, so the molecule can not be polar.

(vi)

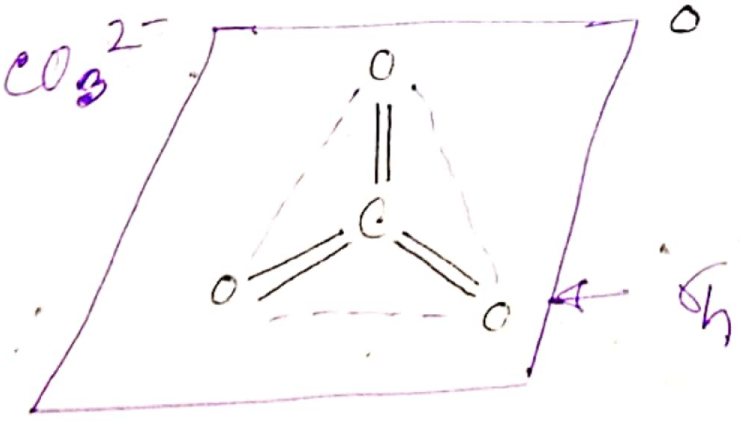


The molecule possess one C_4 axis of rotation.

XeO_3



CO_3^{2-}

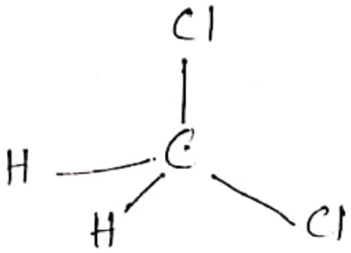


Scanned with CamScanner

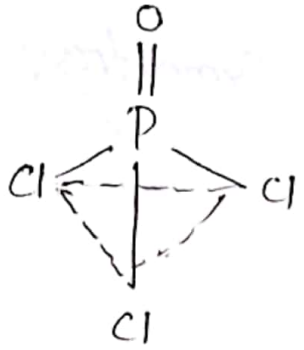
(I) A molecule is not chairal, if it has possesses an improper axis of rotation.

(II) Molecule having σ, i is known as optically inacti

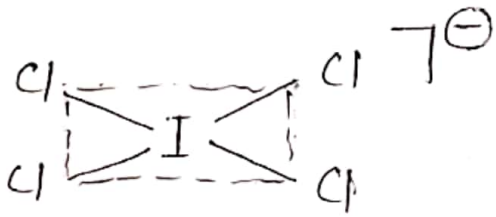
(III) A molecule can not be polar, if it has a center of symmetry.



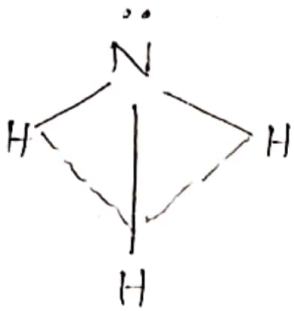
$POCl_3$



ICl_4^-

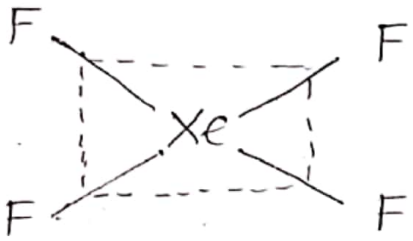


NH_3

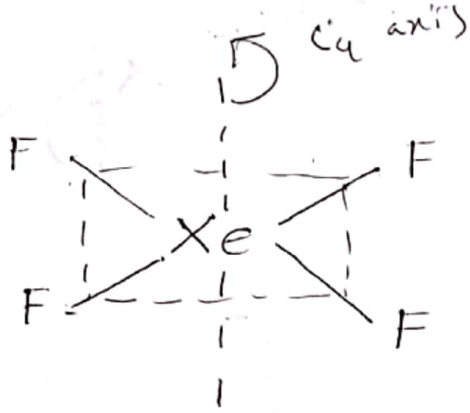


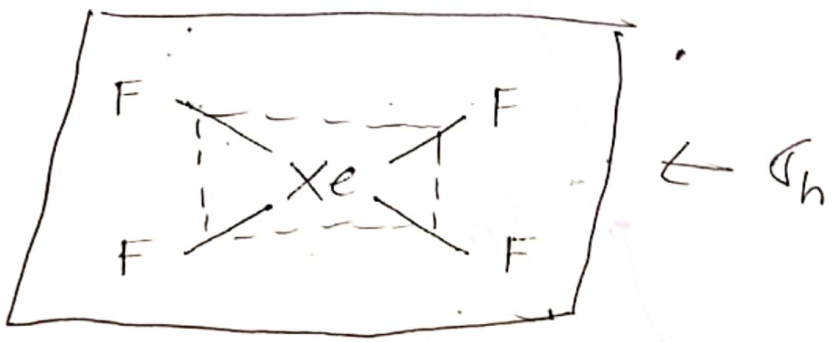
Scanned with CamScanner

XeF4



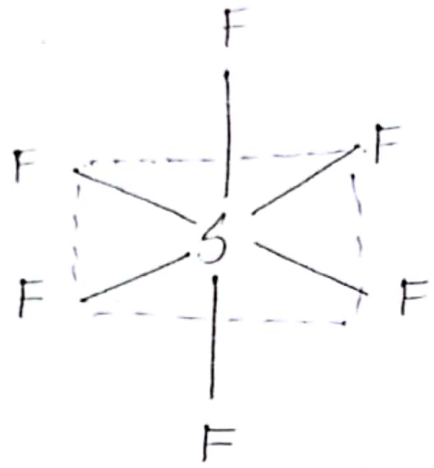
Square planar



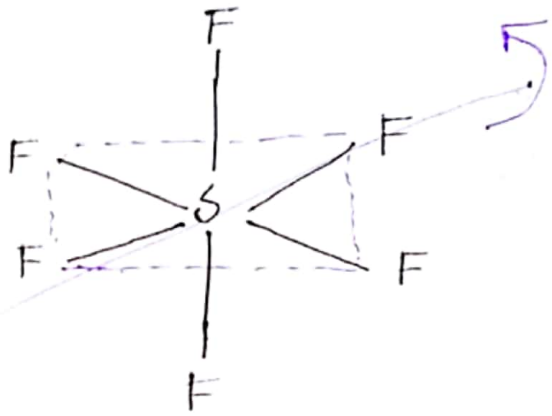


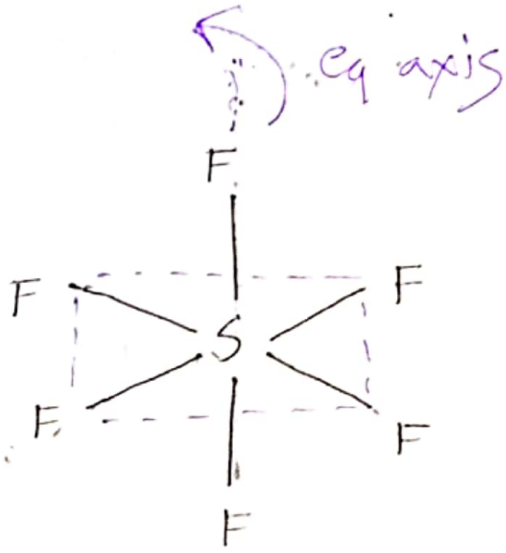
Scanned with CamScanner

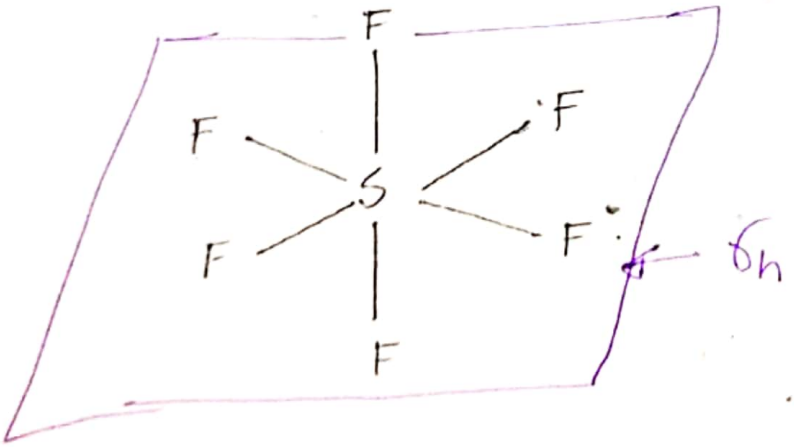
SF_6



Octahedral

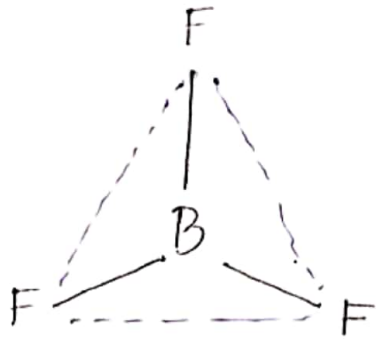




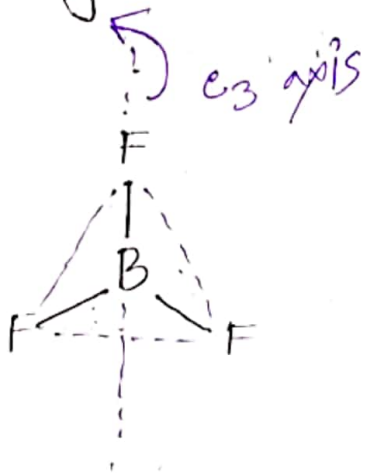


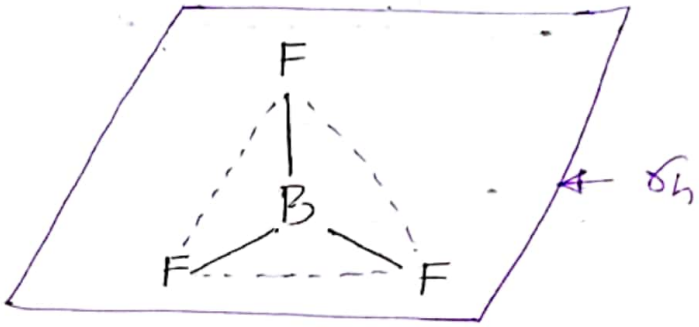
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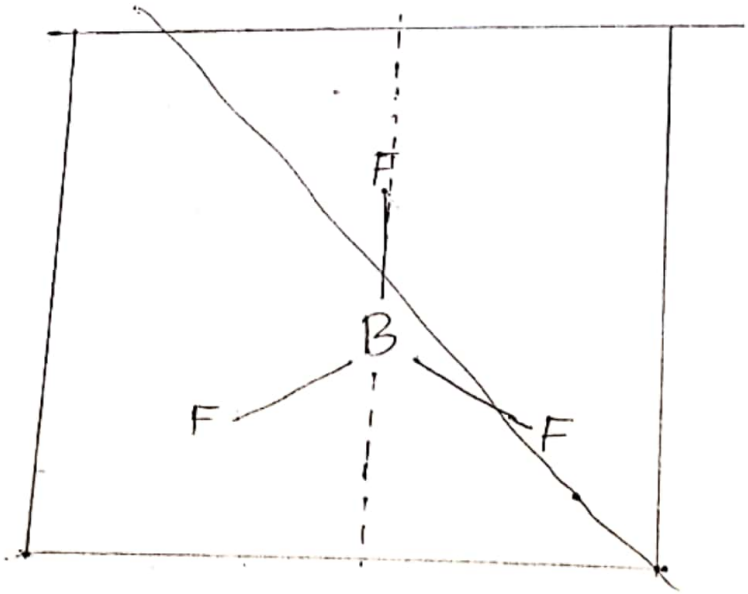
BF_3



Triangular planar

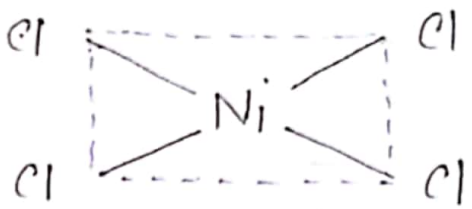






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$[NiCl_4]^{2-}$



square planar

$C_n = ?$

yes, $n = 4$

$4\,\sigma_v$

yes

$4\,C_2 \perp C_4$

yes

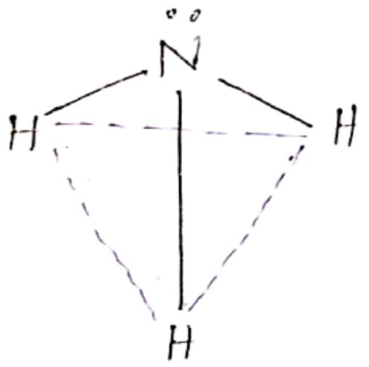
σ_h

yes

D_{4h}

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NH_3



Trigonal pyramidal

Cn = ?
yes, n = 3

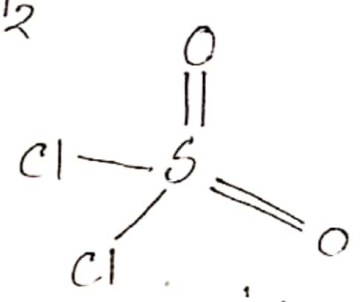
3 σ_v
yes

$3C_2 \perp C_3$

No

C_{3v}

SO_2Cl_2



Tetrahedral

Cn = ?
yes, n = 2

2 σ_v
yes

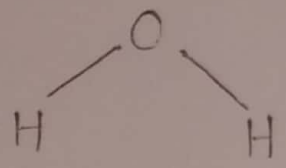
$2C_2 \perp C_2$

No

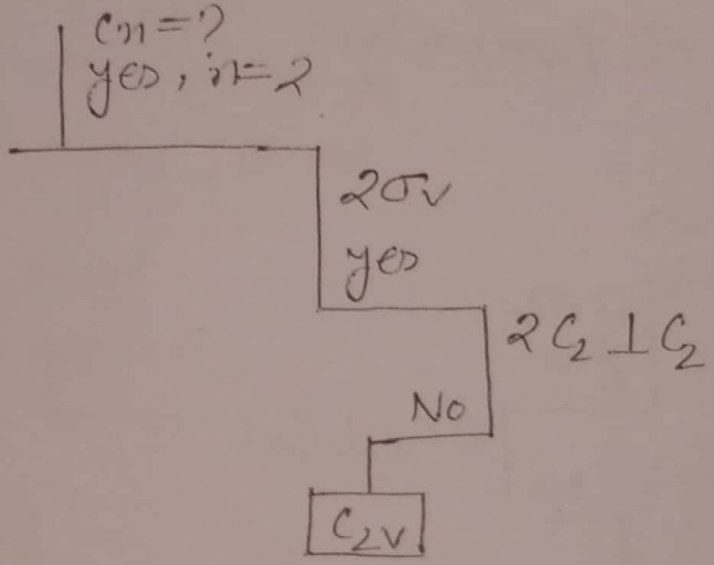
C_{2v}

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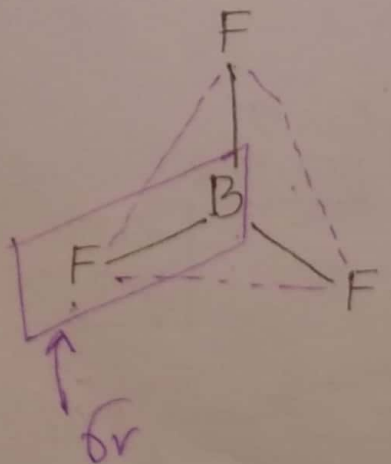
H_2O



V-shape



BF_3



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Cl_2

$Cl-Cl$

linear

Cl_2 is a symmetrical linear molecule. So the point group of Cl_2 is $D_{\infty h}$.

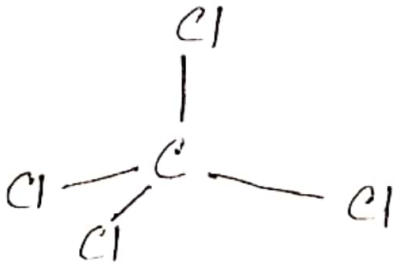
HCl

$H-Cl$

linear

HCl is an unsymmetrical molecule. So the point group of HCl is $C_{\infty v}$.

CCl_4

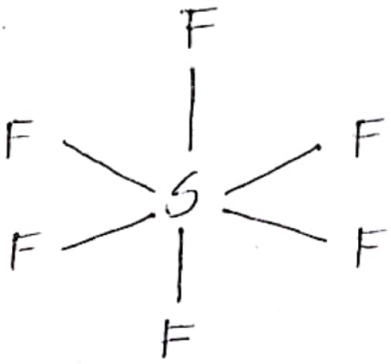


Tetrahedral

CCl_4 is a regular tetrahedral molecule. So the point group is T_d.

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SF_6



Octahedral

SF_6 is a regular octahedral molecule. So the point group is O_h.

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